Assignment 1: ***Interactive Art Storyboarding - "Is Recycling Enough?"***

Name: **Anthony Carey, Janhavi Amit Koparkar, Dharshini Ramesh Babu**

Student ID: **13154494, 25227467, 25006274**

Table of Contents

1. Introduction
2. Project Overview
3. Concept and Thematic Framework

    3.1 SDG Context

    3.2 Critical Sustainability Framework

4. Installation Design

    4.1 Spatial Configuration

    4.2 Projection-Mapped Bottle

    4.3 Physical Recycling Interface

5. Storyboard Development

    5.1 Frame Sequence Overview

    5.2 Escalation and State Transitions

6. Interaction and System Logic

    6.1 Inputs

    6.2 Processing and Algorithm

    6.3 Outputs

    6.4 Feedback Loop

7. Flow Chart Explanation
8. Technology for Realisation

    8.1 Hardware

    8.2 Software

    8.3 Tool Selection Rationale

9. Rules and Algorithmic Structure

10. Theoretical and Module Context

10.1 Cybernetics and Feedback Systems

10.2 Systems Aesthetics

10.3 Behavioural Art

11. Relation to Interaction Design and UX

11.1 Tangible Interaction

11.2 Embodied and Spatial Interaction

11.3 Behavioural Design

12. Intent Statement
13. Conclusion
14. References
15. Appendices

A. Mind Map

B. Storyboard Sketches

C. Flow Chart Diagram

1. Introduction

This project proposes an interactive installation exploring SDG 12: Responsible Consumption and Production. The work examines the imbalance between individual recycling behaviour and large-scale plastic production systems. Rather than presenting recycling as a simple solution, the installation stages sustainability as a cybernetic system in which individual effort is continuously outpaced by systemic production and corporate profit.

The installation combines physical computing, projection mapping, and rule-based algorithmic design to visualise this imbalance spatially and interactively.

2. Concept and Thematic Framework

The core theme of the project is the tension between individual responsibility and structural economic systems. While recycling is often framed as a moral obligation of the individual, global plastic production continues to rise due to industrial and profit-driven systems.

The installation interrogates this contradiction.

Participants insert plastic bottles into a recycling machine and receive a small monetary reward. Simultaneously, a projection-mapped sculptural bottle mounted horizontally on the wall visualises global plastic production and corporate profit increasing automatically over time. As the system escalates, rewards diminish and the projected bottle fills with plastic waste.

The final state questions whether behavioural incentives are sufficient to address systemic environmental crises.

Primary SDG:

- SDG 12: Responsible Consumption and Production

Secondary SDGs:

- SDG 13: Climate Action
- SDG 14: Life Below Water

3. Installation Design

The installation consists of:

- A physical recycling machine with bottle slot and digital reward display
- A large plastic bottle sculpture cut lengthwise and mounted horizontally on the wall
- A digital projector mapping real-time visuals onto the curved interior surface of the bottle
- Sensor-based input detection
- Algorithm-driven visual escalation

Projection mapping transforms the sculptural bottle into both container and screen. Data and ocean imagery are mapped onto its curved surface, reinforcing the conceptual framework by positioning the plastic product itself as the site of accumulation.

4. Interaction and System Logic

The system operates as a feedback loop structured around user input and algorithmic escalation.

Inputs:

- Bottle insertion (physical interaction)
- Simulated global plastic production (automatic increase over time)

Processing:

- Sensor detection of bottle
- Microcontroller logs user input

- Algorithm calculates reward value and system state
- Continuous growth of production variable

Outputs:

- Reward display updates
- Projection mapping updates
- Data counter increases
- Optional sound/light changes

Feedback Loop:

User perceives visual change → user continues interaction → system escalates.

The system intentionally ensures that production increases at a faster rate than recycling input, creating visible imbalance.

5. Rules and Algorithm

The installation operates through rule-based logic.

Variables:

B = Bottles inserted

P = Global production rate (auto-increasing)

R = Reward value

S = System state

Example logic:

- When bottle inserted:

B = B + 1

- Production increases automatically over time:

$P = P + \text{constant}$

- Reward calculation:

$R = \text{base reward} - (P / \text{scaling factor})$

Conditional states:

- If $R > 5¢ \rightarrow$ Balanced state
- If $R \leq 5¢ \rightarrow$ Imbalance
- If $R = 0 \rightarrow$ Breakdown / glitch

These rules create escalation and state transitions, which are visually communicated through projection mapping.

6. Technology for Realisation

Hardware:

- IR object detection sensor
- Arduino or Raspberry Pi microcontroller
- Digital reward display
- Physical sculptural bottle
- Digital projector

Software:

- Processing or p5.js for generative visuals
- Projection mapping software
- Algorithmic rule engine
- Simulated dataset for global production

These tools were selected to enable real-time interaction, spatial visualisation, and rule-based escalation.

7. Intent Statement

"Is Recycling Enough?" stages sustainability as a feedback system in which individual recycling behaviour is continuously outpaced by systemic production. By rewarding users while simultaneously visualising escalating global plastic production and corporate profit, the installation exposes the imbalance between personal responsibility and structural economic forces. The work invites reflection on whether sustainability has been reduced to behavioural incentives rather than structural transformation.

8. Theoretical and Module Context

This project is grounded in the module's exploration of interactive art, systems thinking, and cybernetic feedback. Rather than designing a conventional user interface, the installation adopts a systems-based approach in which behaviour, algorithm, and environment form a feedback loop.

Cybernetics and Feedback Systems

The structure of the installation is influenced by Norbert Wiener's concept of cybernetics as "control and communication in the animal and the machine." The installation functions as a cybernetic loop:

User action → system processing → visual feedback → behavioural response.

However, unlike a balanced feedback system, the installation introduces asymmetry: global production increases independently of user input. This creates a deliberate imbalance that visualises systemic forces overpowering individual action.

The project also draws on Roy Ascott's writings on behavioural art, where the artwork is not an object but a responsive system. Here, the artwork exists as an evolving state machine rather than a static sculpture.

Systems Aesthetics

The installation is also informed by Jack Burnham's concept of "systems aesthetics," which shifts focus from isolated art objects to interrelated systems. Rather than presenting plastic waste as a static visual problem, the installation frames sustainability as a dynamic system of inputs, outputs, and escalation.

The giant projection-mapped bottle is not merely symbolic; it functions as a visualised system container. Data, behaviour, and accumulation become spatially embodied within the sculptural form.

9. Relation to Interaction Design / UX

Although positioned as an art installation, the project engages directly with core areas of Interaction Design and UX.

- **Tangible Interaction**

The bottle insertion mechanic draws from tangible interaction design, where physical manipulation produces digital change. The recycling machine is intentionally simple to emphasise embodied input rather than screen-based interaction.

This reflects the shift in UX from purely graphical interfaces to physical computing and embodied systems.

- **Embodiment and Spatial Interaction**

Projection mapping onto a curved bottle surface moves the experience into spatial storytelling. The interaction is not confined to a flat display; it is distributed across architecture, sculpture, and projection.

The user experiences system escalation physically in space, reinforcing the environmental theme.

- **Feedback and Behavioural Design**

The diminishing reward system engages behavioural design principles. Initially, the user receives positive reinforcement. As the system escalates, this reinforcement weakens and collapses.

This explores:

- Incentive structures
- Behaviour modification
- Transactional sustainability

Rather than designing for optimisation, the project intentionally destabilises the reward loop to provoke reflection.

- **Projection Mapping as Narrative Medium**

Projection mapping is used not just as spectacle but as spatial narrative device. The curved surface of the bottle distorts data and imagery, reinforcing accumulation and containment.

This aligns with contemporary immersive installation practices, where surface and content are inseparable.

10. Influence of Reading Material

The project was influenced by key theoretical texts discussed in the module:

Norbert Wiener – Cybernetics

Provided the conceptual basis for understanding feedback loops and system control.

Roy Ascott – Behavioural Art

Influenced the idea of the artwork as responsive process rather than static object.

Jack Burnham – Systems Aesthetics

Framed the shift from object-based art to systems-based art practice.

These readings helped guide the transition from a simple recycling game concept to a critical systems installation.

Responsible Consumption & Production

Interaction Model

- Deloged Reaction
- Non - Linear Response
- Positive vs negative Feedback.
- continuous Feedback

Overproduction

- Mass Manufacturing
- Plastic Dependency
- Fast Consumption
- Resource Extraction

Individual Vs. System.

- Personal Responsibility
- Structural Responsibility
- Behaviour change
- System imbalance.

Feedback

- Action → Reaction
- Behaviour Reinforcement
- Positive Feedback
- System Escalation
- Imbalance Amplification.

Waste

- Ocean Pollution
- Landfill overflow
- Microplastics
- Recycling limits

Capitalism & Profit

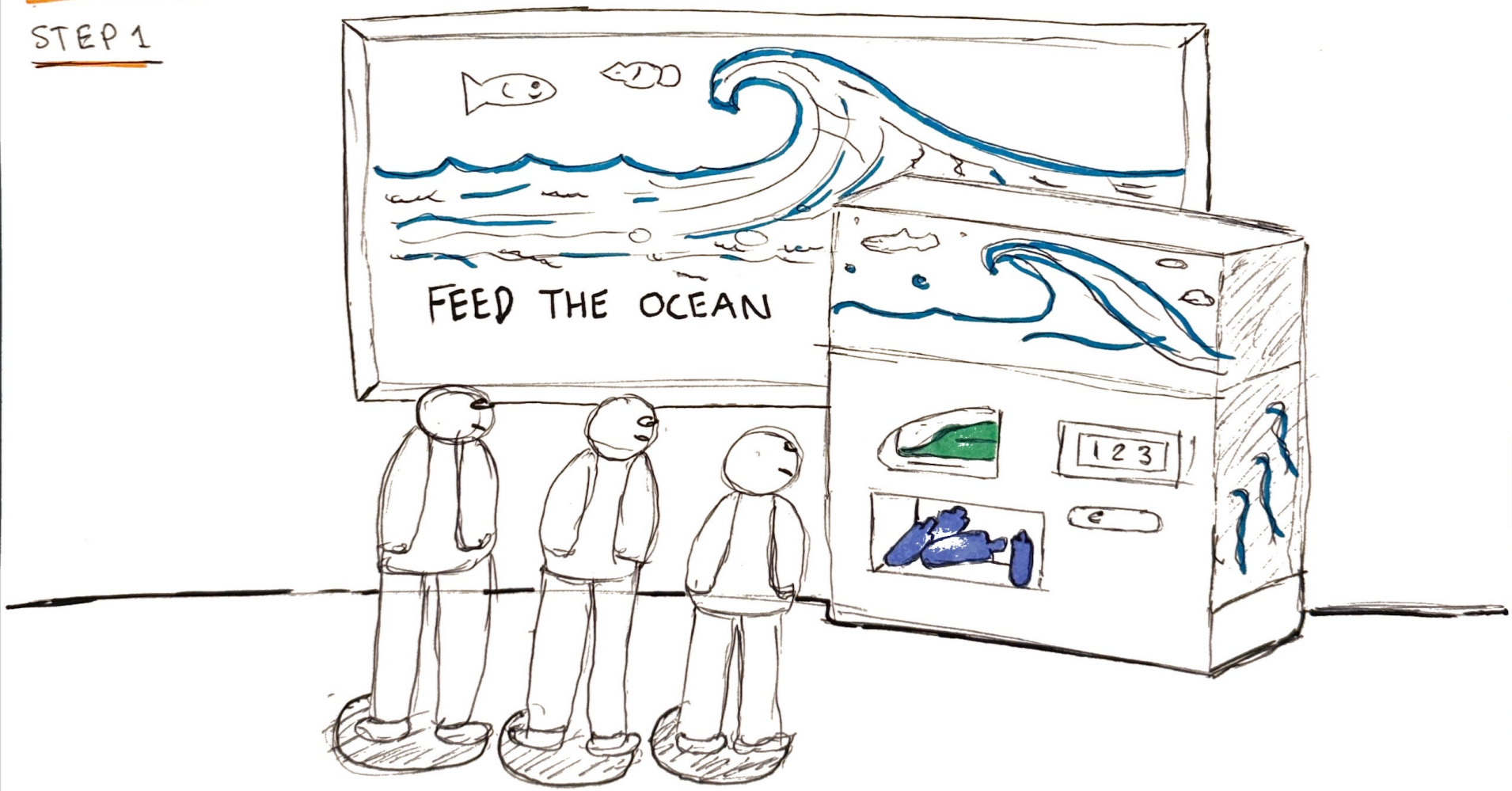
- Corporate gain
- Bottled Water industry
- incentive Systems
- Externalised Cost

Technology

- Object detection
- Touch input
- Live data feed
- Data Comparison

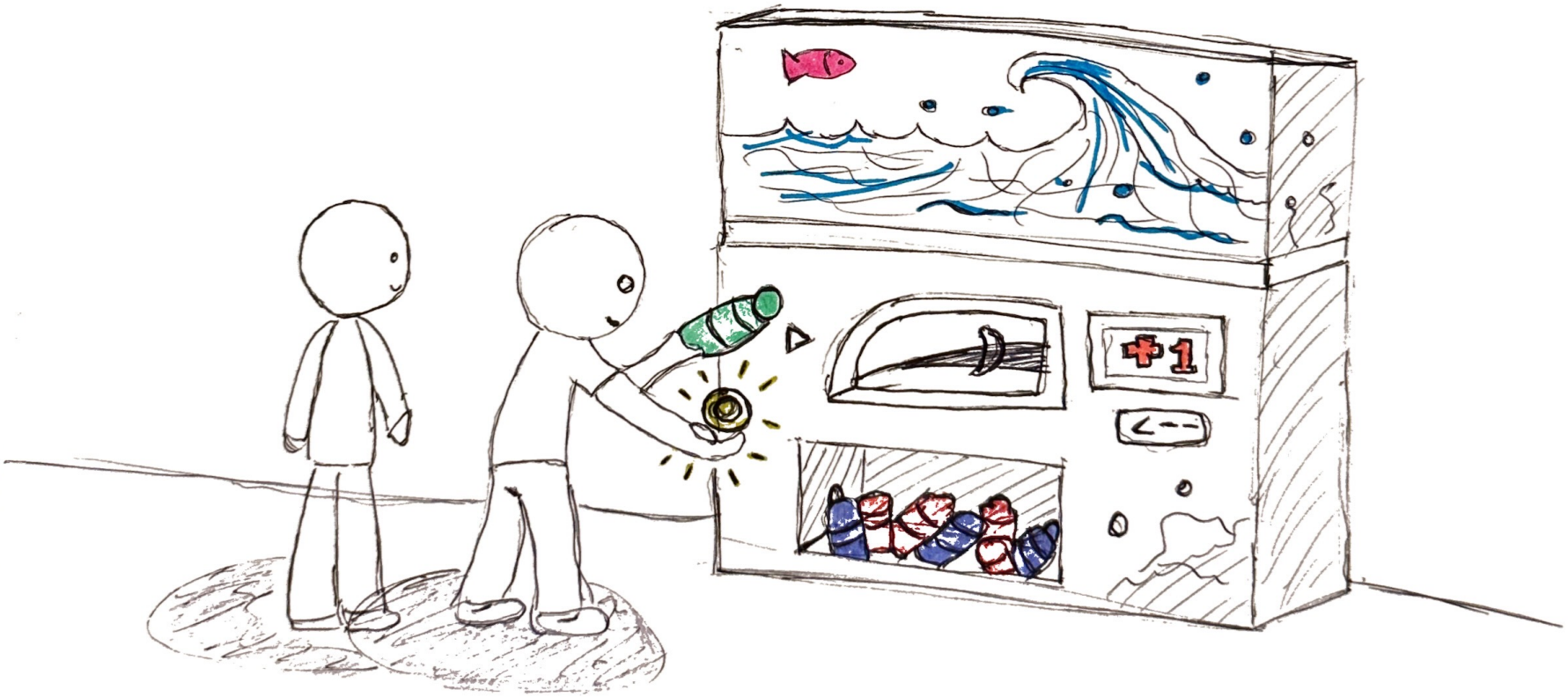
FRAME 1

STEP 1



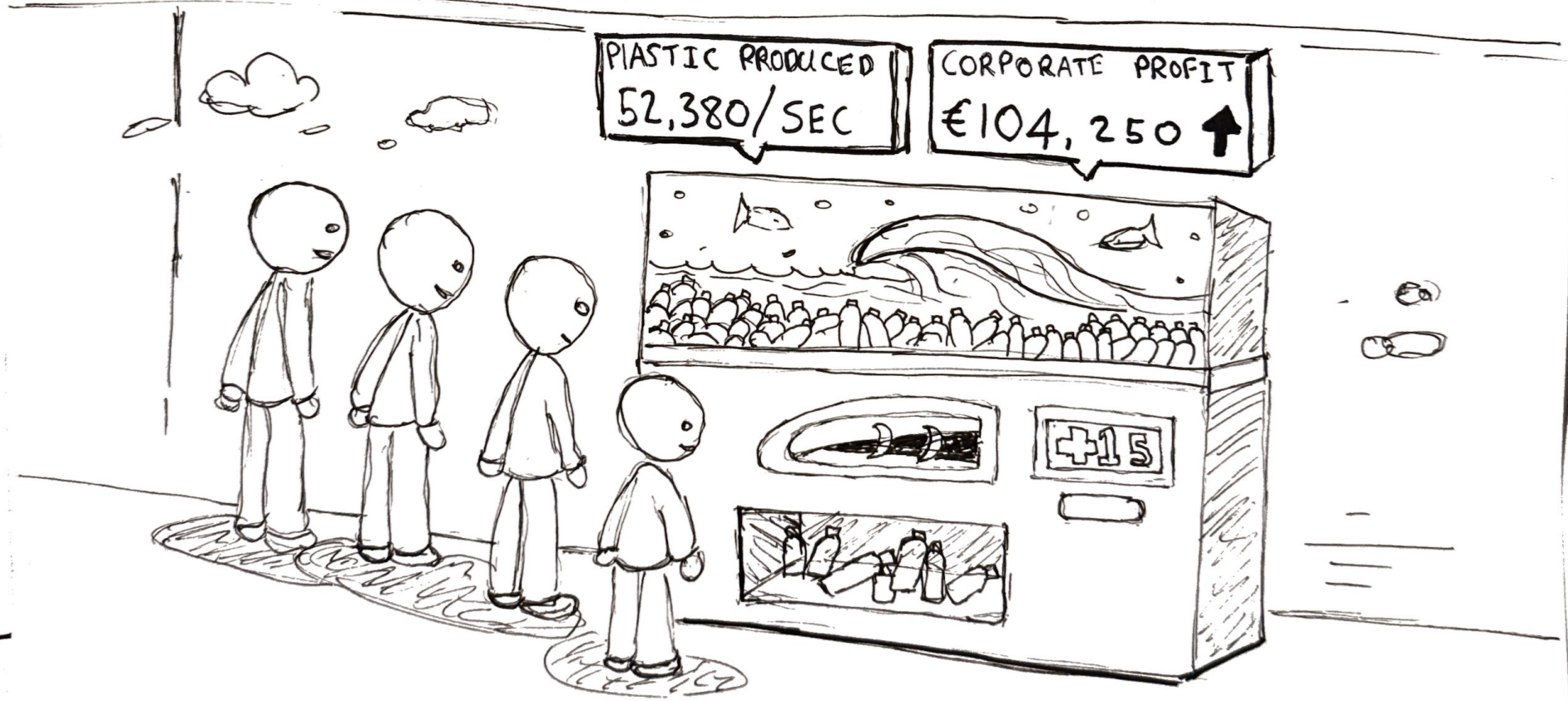
FRAME 2

STEP 2



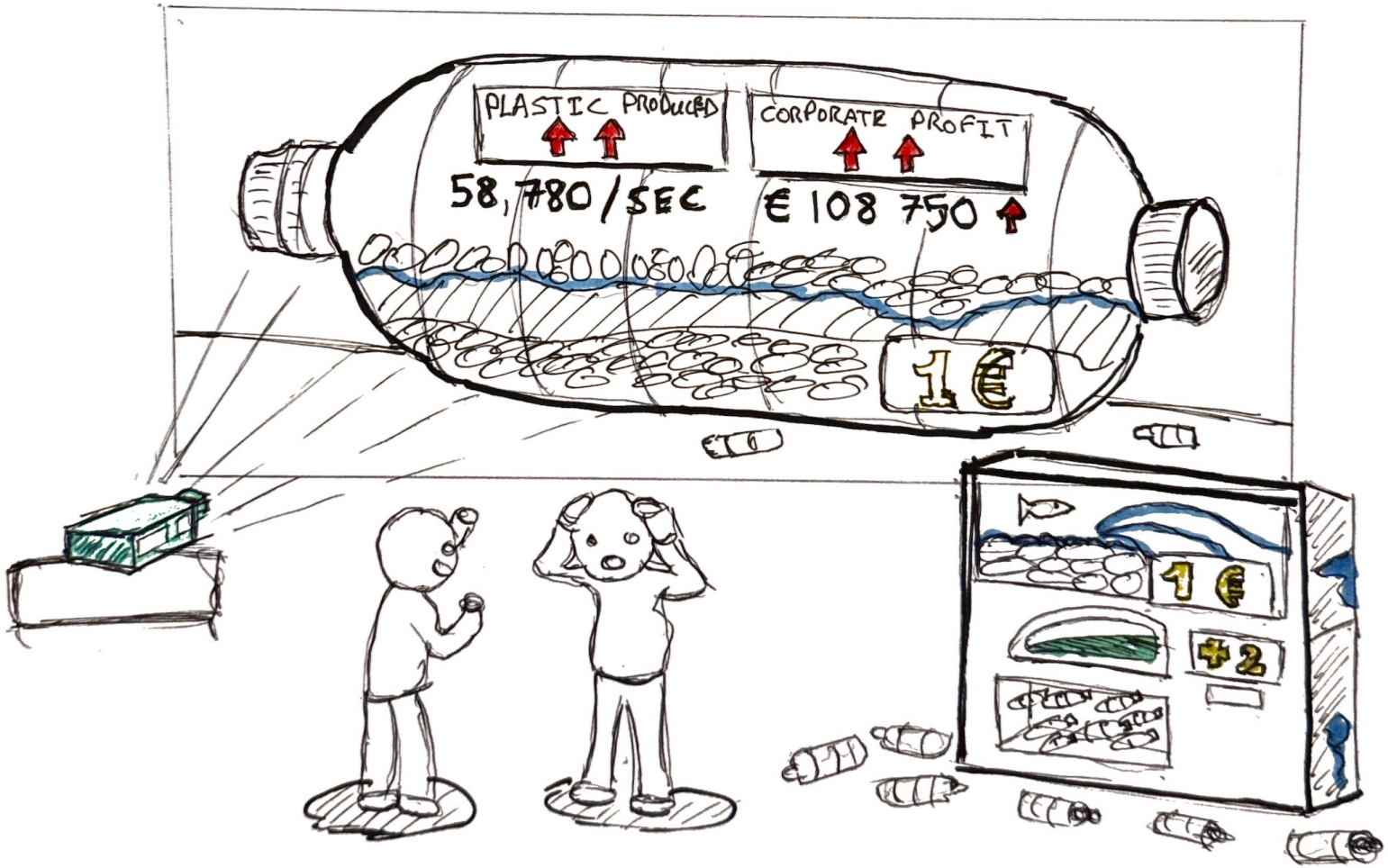
FRAME 3

STEP 3



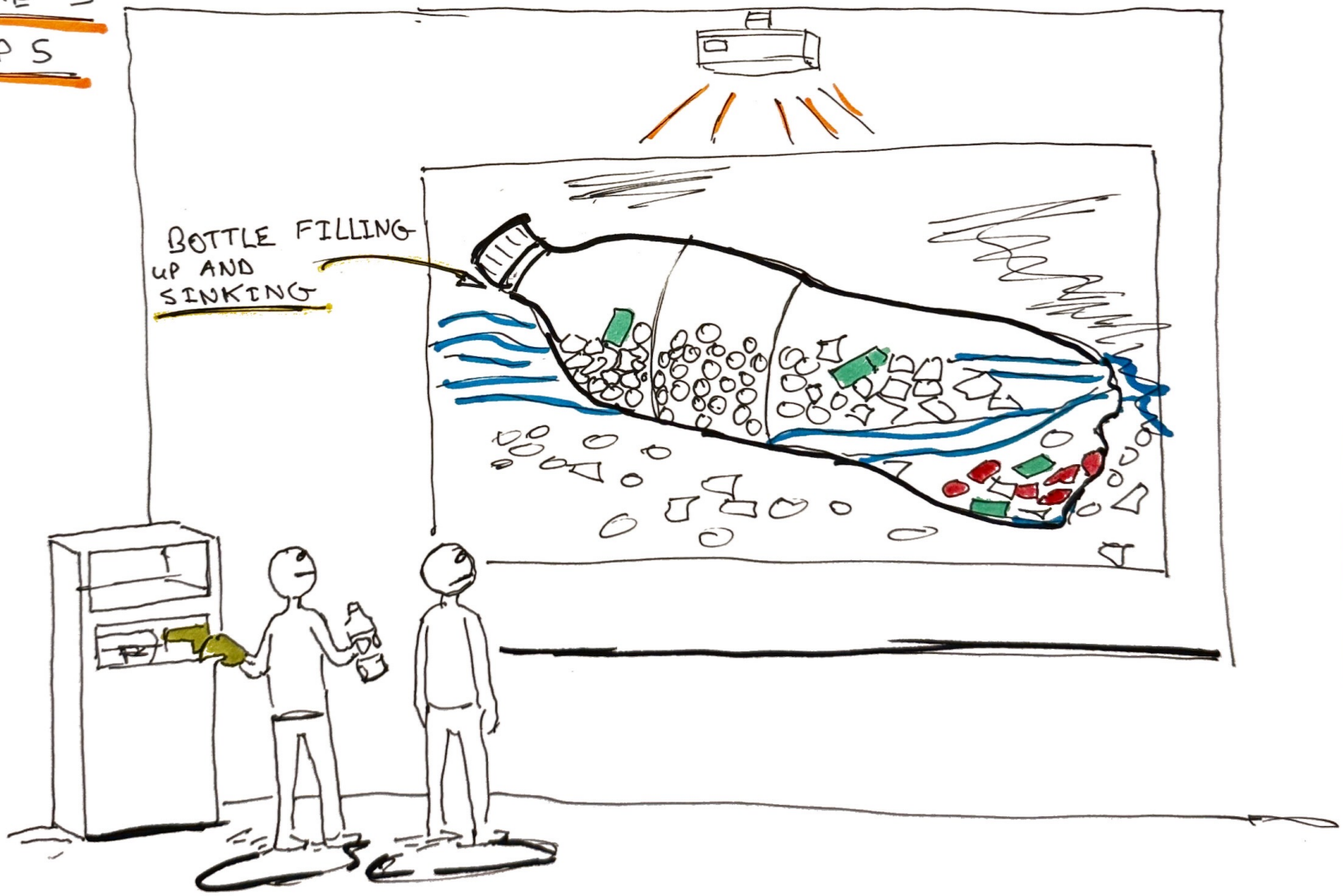
FRAME 4

STEP 4



FRAME 5

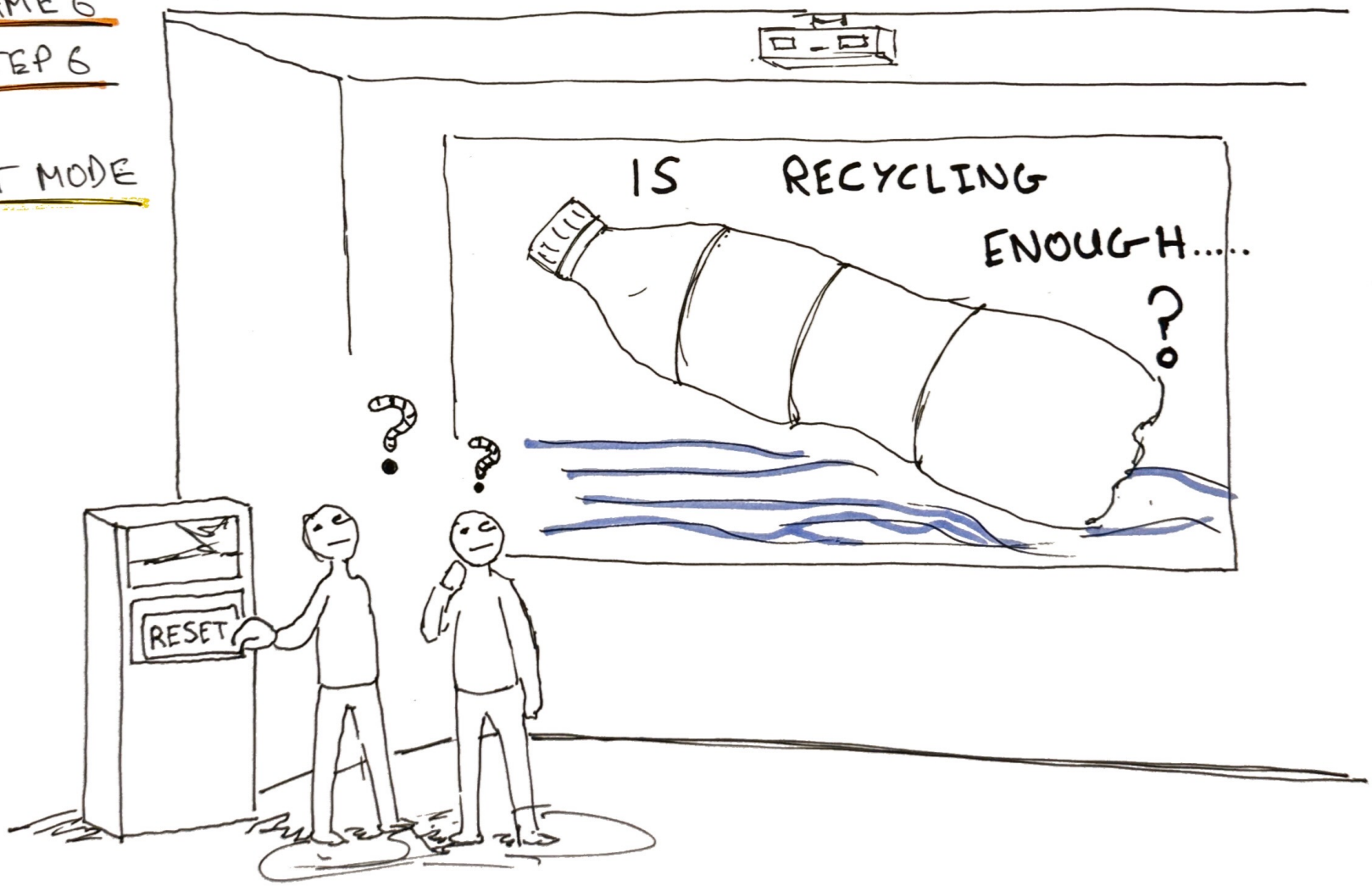
STEP 5



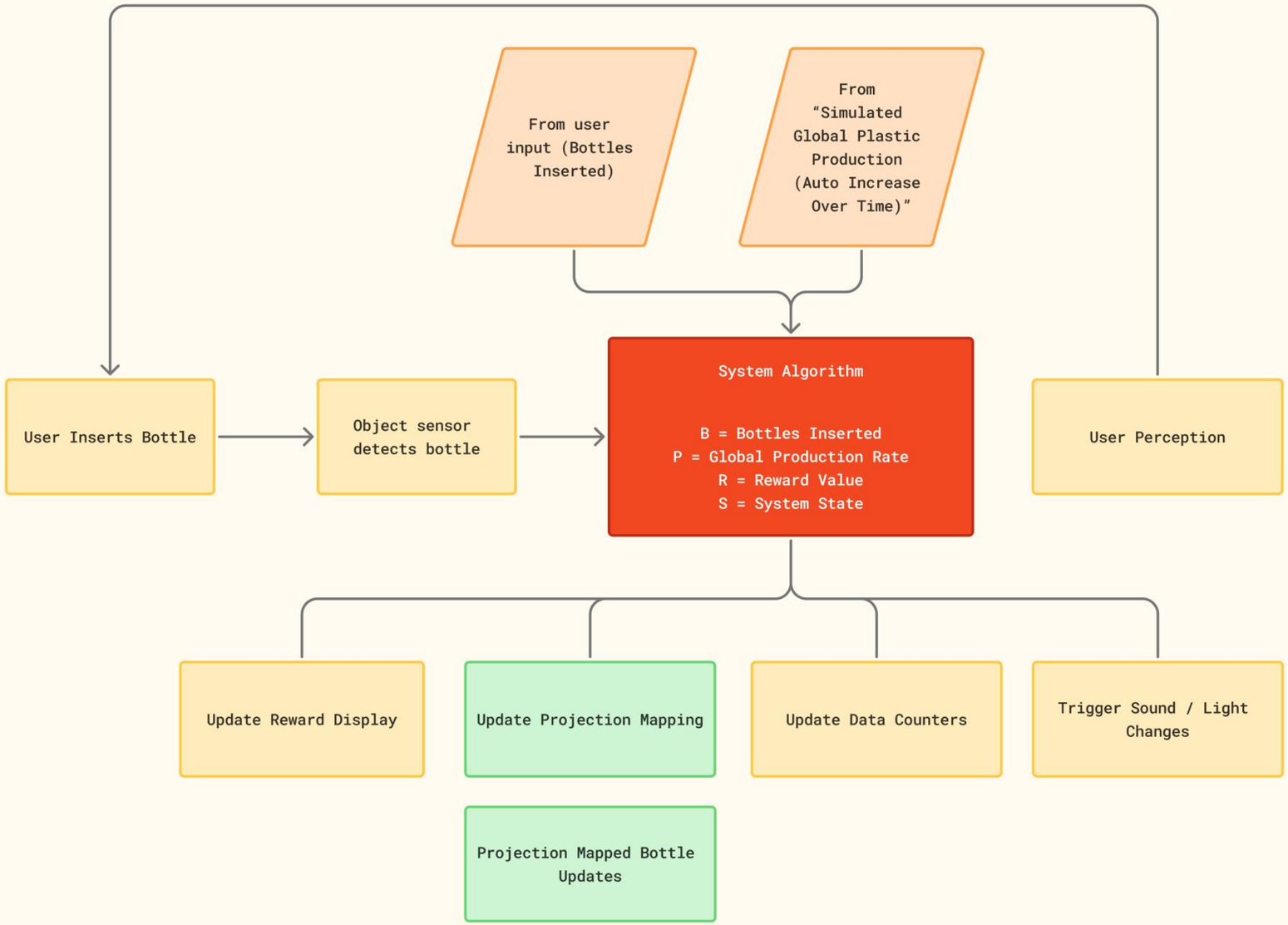
FRAME 6

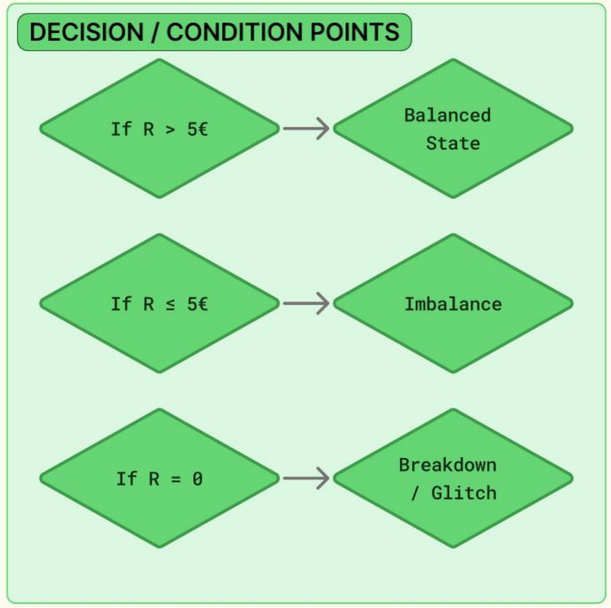
STEP 6

Reset MODE



FLOW CHART


From user input (Bottles Inserted)
From "Simulated Global Plastic Production (Auto Increase Over Time)"
User Inserts Bottle
Object sensor detects bottle
System Algorithm
B = Bottles Inserted
P = Global Production Rate
R = Reward Value
S = System State
User Perception
Update Reward Display
Update Projection Mapping
Update Data Counters
Trigger Sound / Light Changes
Projection Mapped Bottle Updates


DECISION / CONDITION POINTS
If R > 5€
Balanced State
If R ≤ 5€
Imbalance
If R = 0
Breakdown / Glitch

THE OCEAN PAYS YOU BACK

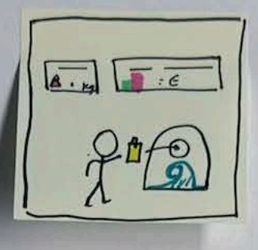
PLASTIC PRODUCED THIS MINUTE

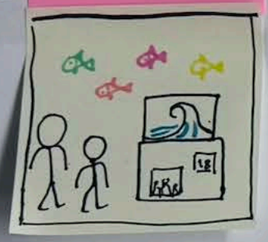
STEP 1

A group of people stand in front of a large ocean themed recycling machine in a gallery space. The machine has a wave illustration on the top display and a bottle slot on the front. Fish and simple underwater lines suggest the ocean theme. The children are observing the installation with curiosity. The system is in a neutral state, waiting for interaction.

• STORYBOARD •

(The Actions)





• THE PROCESS •

(Rules of the Interaction)

• TECHNOLOGY •

(The Systems that we can use)

Interactive Media - week 3:

- Do the readings! from W2 and W1

Interactive Art Piece:

Constraints (fixed / Variable)

Rules

Process

└ USER → ACTION → SYSTEM → NEW STATE.

what triggers change?

*TIME:

what would you design first?

Visuals, Input, Environment, feedback.

└ → Part of the system/ Process.

Interactive Art Storyboarding

└ Assignment.

- Sketches of what happens when the user does this, what happens next?
- How does the system process the input.
- How the information is processed.
- Critical Parts
  - └ engageing for kids
    - └ feedback loop (what makes it worthwhile to put in, (feeling of reward).
- Blake Fall Conroy.